## Trigonometric ratios from points

**1**

**a** $r = 17$ units

**b**
- i $\frac{8}{17}$
- ii $\frac{15}{17}$
- iii $\frac{8}{15}$
- iv $\frac{17}{8}$
- v $\frac{17}{15}$
- vi $\frac{15}{8}$

**2**

**a** $r = 25$ units

**b**
- i $-\frac{24}{25}$
- ii $-\frac{7}{25}$
- iii $\frac{24}{7}$
- iv $-\frac{25}{24}$
- v $-\frac{25}{7}$
- vi $\frac{7}{24}$

**3**

**a** $r = 25$ units

**b**
- i $\frac{24}{25}$
- ii $-\frac{7}{25}$
- iii $-\frac{24}{7}$
- iv $\frac{25}{24}$
- v $-\frac{25}{7}$
- vi $-\frac{7}{24}$

**4**

**a**
- i $\frac{4}{5}$
- ii $\frac{3}{5}$
- iii $\frac{4}{3}$

**b**
- i $\frac{20}{29}$
- ii $-\frac{21}{29}$
- iii $-\frac{20}{21}$

**c**
- i $-\frac{21}{29}$
- ii $\frac{20}{29}$
- iii $-\frac{21}{20}$

## Identify quadrants

**5** $\frac{-\sqrt{15}}{8}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

**6**

**a** Quadrant 2

**b** Quadrant 3

**c** Quadrant 2

**d** Quadrant 1

**e** Quadrant 3 and 4

**f** Quadrant 1 and 4

**g** Quadrant 1 and 3

**h** None

**i** Quadrant 1

**j** Quadrant 4

**k** Quadrant 1

**l** Quadrant 3

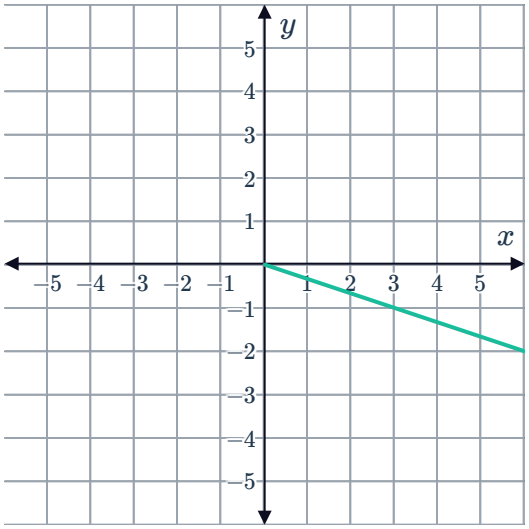
**7**

**a** Quadrant 1 **b** Quadrant 4 **c** Quadrant 3 **d** Quadrant 2

## Equation of the terminal side of an angle

**8**

**a**




**b** $3,\ 1,\ \sqrt{10}$

**c**

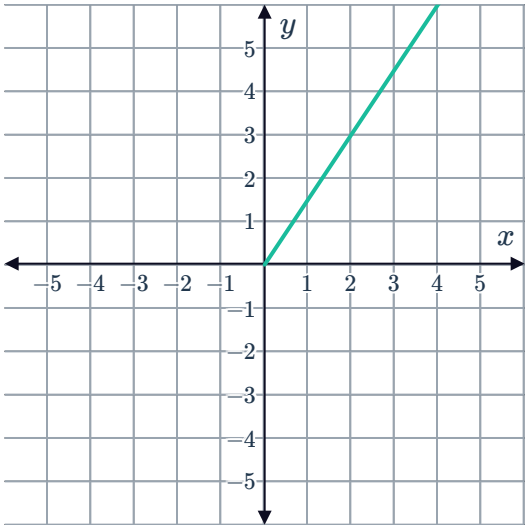
i $-\frac{\sqrt{10}}{10}$ ii $\frac{3\sqrt{10}}{10}$ iii $-\frac{1}{3}$ iv $-\sqrt{10}$

v $\frac{\sqrt{10}}{3}$ vi $-3$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

**9**

**a**




**b** $2,\ 3,\ \sqrt{13}$

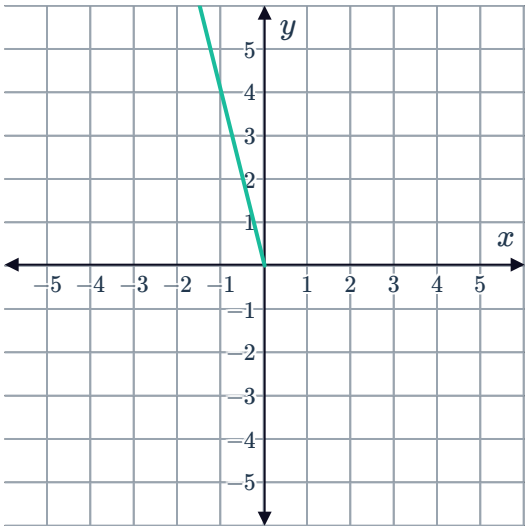
**c**

i $\dfrac{3\sqrt{13}}{13}$ ii $\dfrac{2\sqrt{13}}{13}$ iii $\dfrac{3}{2}$ iv $\dfrac{\sqrt{13}}{3}$

v $\dfrac{\sqrt{13}}{2}$ vi $\dfrac{2}{3}$

**10**

**a**




**b** $4,\ 1,\ \sqrt{17}$

**c**

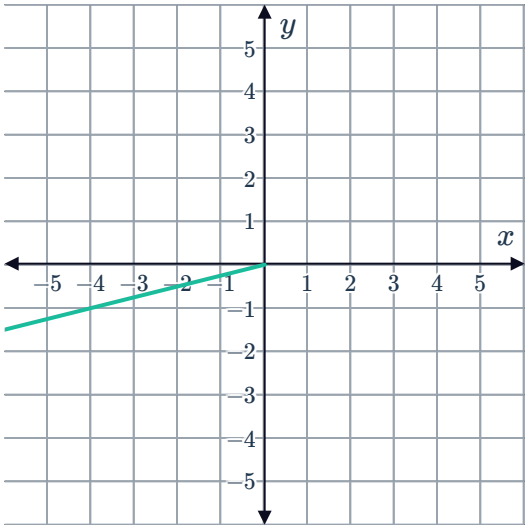
i $\dfrac{4\sqrt{17}}{17}$ ii $-\dfrac{\sqrt{17}}{17}$ iii $-4$ iv $\dfrac{\sqrt{17}}{4}$

v $-\sqrt{17}$ vi $-\dfrac{1}{4}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

**11**

**a**




**b** $4,\ 1,\ \sqrt{17}$

**c**

**i** $-\dfrac{\sqrt{17}}{17}$ **ii** $-\dfrac{4\sqrt{17}}{17}$ **iii** $\dfrac{1}{4}$ **iv** $-\sqrt{17}$

**v** $-\dfrac{\sqrt{17}}{4}$ **vi** $4$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647